

Philip (Phi) Bui

📧 phbui.github.io | ✉ bilphui@gmail.com |  linkedin.com/in/phi-bui |  github.com/phbui

EDUCATION

- Ph.D. in Urban Systems, New York University (NYU), New York, NY** Sep 2026 - Present
- Studying honest uncertainty in urban infrastructure sensing.
 - Doctoral aims: quantify measurement uncertainty, standardize it, and mitigate it with uncertainty-quantified multi-modal data fusion.
 - Advised by Dr. Debra Laefer, Urban Modeling Group.
- M.S. in Computer Science: Human-Robot Interaction, Tufts University, Medford, MA** Sep 2024 - May 2025
- Graduated with a 3.8/4.0 GPA.
- B.S. in Computer Science, Worcester Polytechnic Institute (WPI), Worcester, MA** Aug 2021 - Dec 2023
- Graduated with a 3.8/4.0 GPA and distinction.

HONORS & AWARDS

- Tandon School of Engineering, NYU, New York, NY** 2026
- School of Engineering Ph.D. Fellowship, \$95,119.
- Department of Environmental Protection, NYC, New York, NY** 2026
- Cyvl's winning submission to the Environmental Technology Lab (ETL) Challenge, selected to the 2026 cohort.
- Jonathan M. Tisch College of Civic Life, Tufts University, Medford, MA** 2025
- Inducted into the Honos Civicus Society for civic impact through Untitled LLC.
- School of Arts and Sciences, WPI, Worcester, MA** 2021 - 2023
- Inducted into Upsilon Pi Epsilon (Computer Science Honor Society).

PUBLICATIONS & PRESENTATIONS

- State-Wise Constrained Policy Shaping for Zero-Shot Runtime Behavior Steering, Tufts University** 2026
- Second author, with Thomas Howell, Robert McPherson, and Dr. Vasanth Sarathy.
 - Peer-reviewed workshop paper. Presented at the AAAI-26 AIGOV workshop in Singapore.
 - Developed a post-training supervisor enforcing safety norms on DQN agents in HighwayEnv; reduced collisions by 97% in-distribution and 99% zero-shot.
 - Full-paper version, State-Wise Constrained Policy Shaping for Runtime Normative Supervision, under review at the AAAI-27 Main Technical Track (advanced to Phase 2, September 2026).
- Scoring Rules Certify Reporting, Not Seeking: A Dilemma for Level-Based Rewards in Sequential, Embodied Perception** 2026
- Sole-author manuscript showing that level-based strictly proper scoring rules certify honest reporting but not honest information-seeking, holding for every strictly proper binary rule via Savage's representation.
 - Presented as a poster at the IROS-26 Full-Shift Robot Co-Workers workshop (non-archival) in Pittsburgh.
- GailBot: A PyPI-Distributed Transcription Pipeline for Conversation-Analytic Research, Tufts University** 2026
- Lab demonstration with Dr. Saul Albert, Dr. Umair Muhammad, and Dr. J.P. de Ruiter at the ICCA-26 workshop at University of Alberta: "Using AI within CA Workflows" (facilitated by Dr. Saul Albert and Dr. Lucas Seuren).

RESEARCH EXPERIENCE & PROJECTS

- Regression to the Mean in Municipal Pavement Condition Panels: Why Year-over-Year Change Overstates Condition-Dependent Deterioration, Independent Research** Aug 2026
- Sole-author manuscript quantifying measurement error and regression to the mean in municipal pavement-condition survey panels from multi-epoch public archives (New York City, San Francisco). In the New York City panel, decomposes rating variance into rater error (~15%) and non-persistent occasion effects (~38–56%). Manuscript in preparation.

Do Agents Disclose What the Data Cannot Support? A Live-Data Benchmark for Abstention, Measurement-Uncertainty Caveats, and Missingness Disclosure over Municipal Open Data, Independent Research

Aug 2026

- Sole-author manuscript presenting a 596-question live-data benchmark over a municipal open-data API,
- Headline finding: the evaluated production model (Claude Haiku 4.5) carries a documented reliability caveat into 100%, yet discloses a discoverable data defect in 0 of 108 eligible answers.
- Benchmark, dataset, and episode logs publicly released. Code and data: github.com/phbui/civic-honesty-benchmark.

Research Intern, AI Futures Collab, WPI, Worcester, MA

May 2025 - Sep 2025

- With Dr. Yunus Telli. Investigated human/AI interfacing and literacy, evaluating phenomenology, affordances, AI interactions, transparency and ethical implications of AI-driven media.

Swarm Robotics AI Research Intern, Air Force Research Laboratory, Eglin Air Force Base, FL

May - Aug 2025

- With Dr. Scott Nivison. Designed heterogeneous multi-agent reinforcement-learning algorithms for swarm coordination and decentralized decision-making under communication constraints. (Classified; details omitted.)

Graduate Research Assistant, Human Interaction Lab, Tufts University, Medford, MA

Jan - May 2025

- With Dr. Saul Albert, Dr. Umair Muhammad, Dr. J.P. de Ruiter.
- Led the back-end overhaul of **GailBot**, a PyPI-distributed transcription pipeline for Jeffersonian conversation-analytic notation, built atop off-the-shelf speech-to-text (Deepgram, Watson, Google, OpenAI).
- Engineered REST APIs, designed the PostgreSQL schema, implemented full CRUD in React; authored cross-platform deployment docs and resolved critical Linux/macOS regressions.

LLM Enhanced Captioning for Neurodivergent Users, Tufts University

Jan - May 2025

- Co-developed a real-time Chromium extension integrating a quantized Mistral-7B model over WebSocket; enriched captions with sentiment and intent, achieved <200 ms latency & >95% word accuracy; two pilots (n=8, n=3) reported 4.5/5 clarity.

LLMs Can Be Judgy Too: Fine-Tuning LLMs for Character-Driven RL, Tufts University

Dec 2024 - May 2025

- Built a PPO pipeline using a fine-tuned LLM as a reward model for facial-AU generation; across 30 scenarios, LLM-only achieved 3.12/5 mean accuracy, the hybrid approach matched at 3.07/5 with 32% less variance, and halved manual labeling effort.

Computer Vision-Guided Sphero Swarm Navigation Framework, Tufts University

Sep - Dec 2024

- Integrated overhead-camera Monte Carlo localization (MCL), probabilistic roadmap (PRM) path planning, and a Bluetooth Low Energy (BLE) microservice layer to enable scalable, real-time collision-free coordination of Sphero Bolt swarms.

Evaluating Preferences in Robot Control Systems, Tufts University

Sep - Dec 2024

- Ran a human-robot interaction (HRI) study (n=11) comparing manual stick control vs. autonomous MCL/PRM navigation on obstacle courses; manual was 30% faster with slightly fewer collisions and preferred by 64% of participants.

AI Platform for Low-Income Housing Development for City of Worcester, MA, WPI Major Qualifying Project (MQP)

Aug - Dec 2023

- With Dr. Daniel Treku, Dr. Matthew Ahrens, Dr. Wilson Wong, Prof. Robert Sarnie, Prof. John Galante, Prof. Suzanne LePage, and Prof. Sara Saberi. Built all of the platform's software. Partnered with Worcester's Chief Development Officer to build an AI-assisted land-acquisition and financing pipeline. Integrated GIS layers and open municipal data into a PostgreSQL/Node.js backend and React dashboard.
- Surfaced parcel-level transparency for civic stakeholders. Published records onto blockchain for transparency and accountability. Integrated with the City of Worcester's low-income housing development pipeline.

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer, Cyvl, Somerville, MA

Aug 2025 - Aug 2026

- Designed the AI ingestion pipeline for non-standard municipal engineering documents and diagrams into geometrically-accurate geospatial files; this became the winning submission to NYC DEP's ETL Challenge, earning selection to the 2026 ETL cohort; resulting pilot converted into a multi-million-dollar contract.
- Built a visual embedding pipeline indexing 500k+ high-resolution street-level images per city into a Qdrant vector store, enabling sub-second semantic retrieval over urban infrastructure imagery.
- Extended production AI pipelines processing petabyte-scale, georeferenced imagery and LiDAR point clouds to identify and analyze pavement distress, signs, markings, and above-ground assets across road networks.
- Shipped dozens of features adopted by hundreds of executive-level stakeholders across municipal public-works and transportation departments.
- Partnered with City of Boston leadership to build and maintain Model Context Protocol (MCP) servers, giving non-technical civic staff natural-language access to infrastructure datasets and narrowing the data-access gap between city hall and the agencies that serve residents.
- Investigated calibrated-confidence methods for infrastructure change-detection as an internal feature; internal findings on measurement inconsistency in repeat surveys and on reward design for honest reporting seeded an independent academic research program (see Research Experience & Projects).

Software Engineer, enLabel, Boston, MA

Jan 2024 - Jan 2025

- Refactored hundreds of defects in legacy C#/ASP .NET code; led React/TypeScript UI redesign, improving workflows and user satisfaction.
- Architected a modular React front end with optimized ASP .NET Web API integration; authored end-to-end regression-testing suite to automate API validation.

Website Developer, ProsperOn, Boston, MA

Jun 2023 - Feb 2024

- Built accessible React/TypeScript financial-simulation pages from Figma; refactored Node.js RESTful APIs to modernize legacy code.
- Piloted platform integrations with MassMutual, Mass Fintech Hub, and Boston University; featured in Fintech Sandbox as a leading financial-wellness startup.

Operations Intern, Amazon, Salem, OR

Jun - Aug 2023

- Developed a Node.js mapping tool to track >7,000 items per warehouse; integrated into onboarding training, cutting per-employee costs by 95%.

COMMUNITY SERVICE

Community Team Coordinator, Cyvl, Somerville, MA

Sep 2025 - Aug 2026

- Instructed the "Infrastructure Intelligence" sessions for the Public Innovation Institute x Cyvl Summer Innovation Program, teaching students to work with municipal infrastructure data and tooling.
- Developed system of community outreach to institutions, university organizations, and data communities such as: Yale (PITSCY), NYU (CUSP), Tufts (JumboHacks), BU (Spark), NEU (SUN Lab), WPI (UPE), Olin, BetaNYC (NYC School of Data).
- Collaborated with university organizations to bring lectures, workshops, sponsorships, hackathons, and research opportunities to students.

Co-Founder & Chief Logistics Officer, Untitled LLC, Boston, MA

Dec 2023 - May 2026

- Launched an underground arts collective securing venues, sponsorships, and studio space; produced sold-out fashion shows and performances for thousands of attendees and millions of social-media views.
- Recruited/managed cross-functional teams, handled legal/financial strategy, coordinated security & harm-reduction, and led fundraising for underrepresented artists.